TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	70.6 ± 3.01
Correlation Stability	84.8 ± 1.66
Distribution Consistency	82.2 ± 3.19
Entropy Preservation	77.7 ± 1.73
Mean Z Score Drift	1.1 ± 0.13
Categorical Fidelity	94.3 ± 2.28
Privacy Risk	2.8 ± 0.60
Data Density	100.0 ± 0.00
Coverage Score	72.3 ± 3.41
Feature Accuracy	-87.6 ± 28.34
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	43.635	-1.111	2.5%	OK
TB	3.299	10.766	+7.467	226.4%	Drift
DB	1.486	4.892	+3.406	229.2%	Drift
Alkphos	290.576	728.961	+438.385	150.9%	Drift
Sgpt	80.714	468.908	+388.194	481.0%	Drift
Sgot	109.911	1257.539	+1147.628	1044.1%	Drift
TP	6.483	6.342	-0.142	2.2%	OK
ALB	3.142	2.497	-0.645	20.5%	Warn
A/G Ratio	0.947	0.810	-0.137	14.5%	Warn
Selector	1.286	1.307	+0.020	1.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	70.6 ± 3.01
Correlation Stability (*2)	84.8 ± 1.66
Distribution Consistency (*3)	82.2 ± 3.19
Entropy Preservation (*4)	77.7 ± 1.73
Mean Z Score Drift (*5)	1.1 ± 0.13
Categorical Fidelity (*6)	94.3 ± 2.28
Privacy Risk (*7)	2.8 ± 0.60
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	72.3 ± 3.41

Feature Accuracy (*10)	-87.6 ± 28.34
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6837606837606838	0.6846153846153846	-0.0017094017094017144	
F1	0.7971299319768493	0.8006540771940596	0.7983183092737792	0.0011883772969298102	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6897435897435897	0.6820512820512821	0.7059829059829059	0.016239316239316248	
F1	0.7901675273822802	0.7865366735421713	0.8012187360402493	0.011051208657969088	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743589	0.4692307692307692	0.4598290598290598	-0.09914529914529915	
F1	0.5587432119380917	0.4104546213814186	0.4068702282049316	-0.15187298373316008	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6564102564102565	0.6538461538461538	0.6623931623931625	0.005982905982905973	
F1	0.7603804390590435	0.7632207778438274	0.7665453488343754	0.006164909775331584	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7017094017094017	0.7042735042735042	-0.0025641025641025663	
F1	0.8279091624529537	0.8241134856263329	0.8261570436499837	-0.0017521188029700706	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6854700854700855	0.6991452991452991	0.0247863247863248	
F1	0.7821343746683586	0.7918117356133166	0.8006814378859289	0.018547063217570305	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7008547008547008	0.6820512820512821	0.7076923076923076	0.006837606837606825	
F1	0.8057719262028611	0.7943533725239813	0.8123524967574388	0.006580570554577647	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.6803418803418804	0.7034188034188034	0.021367521367521368	
F1	0.7886399467903636	0.7869878965542569	0.8016012298642675	0.012961283073903918	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6965811965811965	0.6897435897435897	-0.016239316239316237	
F1	0.8125405288538398	0.8175575994749089	0.8133506923850208	0.00081016353118082 97	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6957264957264957	0.6641025641025642	0.7059829059829059	0.010256410256410243	
F1	0.8018521613888432	0.7812170222046274	0.814372697961252	0.012520536572408703	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	3.8368230383554405	1.5599340049435257	239	23	30	6.741614762758045	3.672087242048473	1.0348189645961863
2	67	Female	0.9021816897690942	0.3241952488137853	212	47	48	5.480832450676014	2.7972709757793477	0.7626014077131689
3	34	Female	4.405435139486928	3.0582339555334963	246	929	119	5.134919561049138	2.9707504975390036	0.8209823453016991
4	39	Male	2.837892757862072	0.7358593041957947	162	58	102	5.216004954774934	2.157298925070511	0.8
5	4	Male	42.8	19.7	2110	10	4929	9.5	0.9	0.3
6	85	Female	0.4	0.1	2110	2000	10	9.5	0.9	0.3
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	90	Male	75.0	18.3	1896	10	4929	9.5	0.9	2.8
2	90	Male	0.4	0.1	63	1680	4929	2.7	0.9	2.8
3	90	Male	0.7	0.1	63	11	4928	2.7	0.9000261197664184	2.8
4	4	Male	0.4	0.1	99	1680	4929	3.7824903483973333	5.497935360193873	2.7999817155638045
5	4	Male	0.4	0.1	63	1680	4929	2.7	0.9	2.8
6	4	Male	0.4	18.3	1896	10	11	2.7	5.5	2.8
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	4	Male	75.0	19.7	63	10	17	3.0	1.0	2.8
2	90	Male	0.4	0.1	63	10	10	3.0	1.0	0.3034026290840598
3	32	Male	9.135380257986633	5.217884136633622	279	1148	127	5.10354041239367	2.9076410569968916	0.8347155462277226
4	28	Male	3.9336363782930253	1.0684385890264325	65	34	96	3.044228179856988	1.550335568012249	1.1
5	35	Male	6.925725268727691	8.508780493454523	378	124	145	6.907455456813307	3.4536794022057133	0.9409659079086546
6	4	Female	0.4	19.7	2110	10	10	3.0	1.0	0.3
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	37	Male	5.640592797671689	2.039007005861354	294	27	37	6.636262599325208	3.526540762745489	1.0
2	87	Male	0.40010852036843725	0.1	1107	1750	112	2.700011690700361	0.9	0.3042691400610157
3	33	Female	9.113844430022478	7.700105641519579	282	1478	141	5.455265110658792	3.1864791749152515	0.8939094592325734
4	35	Male	2.729780658099493	0.8264601432606127	176	65	106	5.348566216758614	2.386748459600269	0.8801963108303776
5	90	Male	75.0	19.7	2110	2000	4929	9.6	0.9944643638319274	0.3
6	4	Female	75.0	0.1	2110	10	10	2.7	0.9	2.8
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39	Male	9.605598088137427	2.4255503212520235	267	25	38	6.385274728195358	3.266154489447761	0.9214285263066846
2	74	Male	1.3190452636016672	0.8889683109213761	261	60	93	5.797082542321362	2.9192459619654705	0.6835842092205969
3	21	Male	32.59084368944769	18.299586782568916	1534	2000	4905	2.8400665873265574	3.0387572190188732	1.0860444215624532
4	4	Male	32.6	18.3	90	10	4929	2.7	0.9	2.8
5	4	Male	0.40012951506115785	18.3	90	10	11	2.7	0.9	2.8
6	47	Male	0.4	18.3	90	10	11	9.6	5.5	2.8
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	48	Male	13.547959440367233	19.692137961475183	68	19	21	3.0168307520356358	0.9525617583181564	2.8
2	4	Male	0.5	0.1	1896	1680	20	2.8	0.9	0.3000202399214389
3	32	Male	8.606097678563849	4.441264478770273	263	1381	126	5.163797666146745	2.996332737076505	0.9288153000781451
4	14	Male	50.030643385232552	19.697183115019952	63	46	3817	7.150574901000875	2.1032289174654384	2.8
5	90	Male	75.0	19.7	63	1680	4929	9.6	3.283993905593259	0.3
6	53	Male	0.9915302591394086	0.2504110925031486	199	36	36	4.127560407547078	1.5316851572251065	0.3672083528837357
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	85	Male	0.4	0.1	75	10	4929	2.8	0.9	0.3
2	37	Male	0.4	0.1	75	10	4929	2.8	0.9	0.3
3	35	Male	7.19534262062081	4.39530524591858	306	1255	125	5.35259046905498	2.95886272003392	0.9
				4				35	4	
4	35	Male	2.58550356585543	0.54001821964102	160	57	88	5.30419653870921	2.30538548837887	0.81044121830214
			5	91					9	41
5	4	Male	75.0	19.7	75	10	4929	9.5	0.9	0.3
6	85	Female	0.7	0.1	75	10	10	2.8	0.9	0.3
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	90	Male	0.4	0.1	63	10	11	2.7	0.9	2.5
2	90	Male	0.4	0.1	63	10	17	2.7	0.9	2.5
3	34	Male	6.35669452326316	3.42720809231424	253	1134	133	5.3011103695939	2.96879900574424	0.91545623186990
			5	63					04	57
4	4	Male	42.8	19.7	2109	10	11	2.7	0.9	2.5
5	4	Female	42.8	19.7	63	2000	4929	2.7	0.9	0.3
6	4	Female	0.4	0.1	63	10	11	2.7	0.9	2.5
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	4	Male	0.4	0.1	2110	10	4929	2.8	0.9	2.8
2	4	Male	0.4	0.1	2110	10	4929	2.8	0.9	0.3
3	4	Male	0.4	0.1	2110	10	4929	2.8	0.9	0.3
4	25	Male	2.72712672180482	2.84026702219954	63	27	206	3.10143910361954	1.39759601171161	1.36671104270948
			14	65				6	13	53
5	4	Male	75.0	19.7	63	2000	4929	9.5	5.5	0.3
6	4	Female	0.4	0.1	63	10	11	2.8	0.9	2.8
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	85	Male	0.4	0.1	1896	10	10	2.7	5.5	0.3
2	67	Male	0.86061090347017	0.33485489127566	215	42	46	5.64922708734715	2.95336378854197	0.8
			19	403					8	
3	85	Male	75.0	19.7	1896	10	4929	2.7	0.9	2.8
4	85	Male	75.0	19.7	63	10	3337	2.7	0.9	2.8
5	4	Male	75.0	19.7	63	2000	4929	2.7	5.5	2.8
6	85	Female	0.4	0.1	1896	10	10	2.7	5.5	0.3
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3